

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Pseudocode Design

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5- Pseudocode Design	Assessment:	Assessment Tool 1– Pseudocode Design
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	To implement Pseudocode Design for different data science algorithms		
Student Name:	M Poojitha Reddy	Reg. No.:	192472352
Section / Batch:	2024	Date:	26-08-26

Code of Conduct

I M Poojitha Reddy - 192472352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Poojitha Reddy

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. K-MEANS CLUSTERING

QUESTION :

Design pseudocode to implement the K-Means clustering algorithm: initialize centroids, assign observations to the nearest centroid, update centroids, and repeat until convergence.

ALGORITHM

1. Start with the dataset and number of clusters **K**.
2. Randomly initialize **K centroids**.
3. Calculate the distance of each observation from every centroid.
4. Assign each observation to the **nearest centroid**.
5. Recalculate each centroid as the **mean of observations** in that cluster.
6. Repeat steps 3–5 until the centroids do not change.
7. Return the final clusters and centroids.

PSEUDOCODE

BEGIN

Input dataset X and number of clusters K

Initialize K centroids randomly

REPEAT

FOR each observation x in X

Calculate distance from x to every centroid

Assign x to the nearest centroid

END FOR

FOR each cluster k = 1 to K

Calculate mean of observations in cluster k

Update centroid[k]

END FOR

UNTIL centroids do not change

Display final clusters and centroids

END

OUTPUT:

Initial centroids are selected.

Observations are assigned to their nearest centroids.

Centroids are updated repeatedly.

The process continues until the centroids do not change.

Final Clusters:

Cluster 1 → Observations belonging to Cluster 1

Cluster 2 → Observations belonging to Cluster 2

...

Cluster K → Observations belonging to Cluster K

Final centroids are displayed.

K-Means clustering converged successfully.

2. K-MEANS CLUSTERING – CUSTOMER GROUPING

QUESTION

Design pseudocode to group customers into three clusters based on annual income and spending score using K-Means clustering.

ALGORITHM

1. Read the customer dataset.
2. Select **Annual Income** and **Spending Score** as features.
3. Set **K = 3**.
4. Initialize three centroids randomly.
5. Calculate the distance of each customer from the three centroids.
6. Assign each customer to the nearest centroid.
7. Calculate new centroids using the mean of each cluster.
8. Repeat until the centroids do not change.
9. Display the three customer clusters.

PSEUDOCODE

BEGIN

Input customer dataset

Select:

X = Annual Income, Spending Score

Set K = 3

Initialize 3 centroids randomly

REPEAT

FOR each customer

Calculate distance to all 3 centroids

Assign customer to nearest centroid

END FOR

FOR each cluster k = 1 to 3

Calculate mean Annual Income

Calculate mean Spending Score

Update centroid[k]

END FOR

UNTIL centroids do not change

Display Cluster 1

Display Cluster 2

Display Cluster 3

END

OUTPUT:

Customer clusters based on Annual Income and Spending Score:

Cluster 1 → Customers with similar income and spending behavior

Cluster 2 → Customers with similar income and spending behavior

Cluster 3 → Customers with similar income and spending behavior

Final centroids:

Cluster 1 → (Mean Income, Mean Spending Score)

Cluster 2 → (Mean Income, Mean Spending Score)

Cluster 3 → (Mean Income, Mean Spending Score)

Customers are successfully grouped into 3 clusters.

3. K-MEANS CLUSTERING – ELBOW METHOD

QUESTION

Design pseudocode to determine the suitable number of clusters using the Elbow Method and then apply K-Means with the selected number of clusters.

ALGORITHM

1. Input the dataset.
2. Select a range of possible values of **K**.
3. Apply K-Means for each value of **K**.
4. Calculate **WCSS (Within-Cluster Sum of Squares)** for each **K**.
5. Plot the values of **K** against **WCSS**.
6. Identify the **elbow point** where the decrease in **WCSS** becomes small.
7. Select the **K** value at the elbow point.
8. Apply K-Means using the selected **K**.
9. Display the final clusters.

PSEUDOCODE

BEGIN

Input dataset **X**

FOR **K** = 1 to **Kmax**

 Apply K-Means with **K** clusters

 Calculate **WCSS**

 Store **WCSS[K]**

END FOR

Plot **K** versus **WCSS**

Find the elbow point

selected_K = elbow point

Apply K-Means using selected_K

Display final clusters

END

OUTPUT:

WCSS values for different K values are calculated.

K = 1 → WCSS = ...

K = 2 → WCSS = ...

K = 3 → WCSS = ...

K = 4 → WCSS = ...

K = 5 → WCSS = ...

The Elbow Point is found at K = 3.

Selected Number of Clusters = 3

K-Means is applied with K = 3.

Final 3 clusters are obtained successfully.

4. EVALUATION METRICS

QUESTION

Design pseudocode to calculate TP, TN, FP, and FN from actual and predicted binary class labels and subsequently calculate accuracy, precision, recall, and F1-score.

ALGORITHM

1. Initialize **TP, TN, FP and FN** to zero.
2. Compare the actual and predicted labels.
3. Count **TP, TN, FP and FN**.

4. Calculate **Accuracy**.
5. Calculate **Precision**.
6. Calculate **Recall**.
7. Calculate **F1-Score**.
8. Display all evaluation metrics.

FORMULAS

$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$

$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

$\text{F1-Score} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

PSEUDOCODE

BEGIN

Input actual labels Y_actual

Input predicted labels Y_predicted

TP = 0

TN = 0

FP = 0

FN = 0

FOR i = 1 to number of observations

IF Y_actual[i] = 1 AND Y_predicted[i] = 1

TP = TP + 1

ELSE IF Y_actual[i] = 0 AND Y_predicted[i] = 0

TN = TN + 1

ELSE IF Y_actual[i] = 0 AND Y_predicted[i] = 1

FP = FP + 1

ELSE IF Y_actual[i] = 1 AND Y_predicted[i] = 0

FN = FN + 1

END IF

END FOR

$Accuracy = (TP + TN) / (TP + TN + FP + FN)$

$Precision = TP / (TP + FP)$

$Recall = TP / (TP + FN)$

$F1_Score = 2 * Precision * Recall /$
 $(Precision + Recall)$

Display TP, TN, FP, FN

Display Accuracy

Display Precision

Display Recall

Display F1_Score

END

OUTPUT:

Confusion Matrix:

TP = 40

TN = 45

FP = 5

FN = 10

Accuracy = 0.85

Precision = 0.89

Recall = 0.80

F1-Score = 0.84

The classification model is evaluated successfully.

5. MODEL EVALUATION

QUESTION

Design pseudocode for comparing kNN, Decision Tree, Logistic Regression, and K-Means models using appropriate evaluation measures and selecting the most suitable model for a given Data Science problem.

ALGORITHM

1. Input the dataset.
2. Preprocess the dataset.
3. Split the data into training and testing sets for classification models.
4. Train **kNN, Decision Tree and Logistic Regression** models.
5. Generate predictions using the test data.
6. Calculate **Accuracy, Precision, Recall and F1-Score**.
7. Apply **K-Means** for clustering.
8. Evaluate K-Means using **Silhouette Score and WCSS**.
9. Compare the models using appropriate evaluation measures.
10. Select the most suitable model based on the problem and evaluation results.

PSEUDOCODE

BEGIN

Input dataset

Preprocess dataset

Split dataset into training and testing data

Train kNN model

Train Decision Tree model

Train Logistic Regression model

Predict test data using each classification model

Calculate:

Accuracy

Precision

Recall

F1-Score

Apply K-Means clustering

Determine suitable K using Elbow Method

Calculate:

WCSS

Silhouette Score

IF problem is classification THEN

Compare kNN, Decision Tree
and Logistic Regression

Select the model with the best
suitable evaluation metrics

ELSE IF problem is clustering THEN

Select K-Means based on
Silhouette Score and WCSS

END IF

Display selected model
Display evaluation results

END

OUTPUT:

Model Evaluation Results:

kNN:

Accuracy = ...

Precision = ...

Recall = ...

F1-Score = ...

Decision Tree:

Accuracy = ...

Precision = ...

Recall = ...

F1-Score = ...

Logistic Regression:

Accuracy = ...

Precision = ...

Recall = ...

F1-Score = ...

K-Means:

WCSS = ...

Silhouette Score = ...

Selected Model = Model with the best suitable evaluation metrics

The most suitable model is selected successfully.